# Supplementary Material: Reference Experiments on the MCC5-THU Motor Dataset—Protocol Difficulty, Modality Sharing, and Feature Physics

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***Abstract*—This document is supplementary material to "Beyond Closed-Set Accuracy: Constituent-Level Generalization in Compound Motor Fault Diagnosis." It reports reference experiments on the MCC5-THU motor dataset that stand on their own, independently of the constituent-level method developed in the main paper: a comparison of six evaluation protocols under operating-condition shift, a measurement of what a shared multimodal feature space does to the weaker constituent of a compound fault, empirical checks on two order-domain feature-construction parameters, and documentation of release-level properties that affect protocol design.**

## I. Scope

The experiments collected here use a generic window-level pipeline, so the measured gaps reflect protocol difficulty and representation structure rather than modeling choices. They are reported separately from the main paper because they characterize the dataset and its evaluation protocols rather than the constituent-level method, and because they may be useful to other groups working with this release.

### *A. Test Rig and Release Structure*

Table I summarizes the machine, instrumentation, and release organization underlying every experiment in this paper. Two properties drive the protocol design. First, each diagnostic state is recorded once per operating condition, so a run is the only sampling unit that is safely independent; window-level splitting would place segments of one physical recording on both sides of a split. Second, the twelve conditions are a crossed grid of two excitation profiles, two load levels, and three speeds, which makes it possible to hold out a condition either by interpolation or by extrapolation—a distinction quantified in Section I-B.

Three release-level properties were observed while preparing the data and are reported because they materially affect protocol construction. (i) The distributed archive contains 288 runs, whereas the descriptor article reports 282; the surplus is a second `bearing_outer_H` recording at the constant-torque 20 Nm, 3000 rpm condition, time-stamped roughly six weeks after the main acquisition campaign. (ii) Compound-fault directory names abbreviate the second constituent, so that a directory named `bearing_outer_H_and_inner_H` denotes an outer-race and inner-race combination; parsing the labels naively splits one compound into inconsistent constituent sets. (iii) Recordings of the same fault state made on different dates are separable with near-perfect accuracy from the same features used for diagnosis (run-level AUC 0.998–0.999). Property (iii) is the reason the transfer analyses in the main study are bounded by recording-sensitivity and same-date controls rather than being read directly as fault physics.

TABLE I
Test rig, instrumentation, and release structure. Machine and sensor parameters are as documented in the dataset descriptor [1]; the release-structure rows are as observed in the distributed files.

| Property | Value |
|---|---|
| Machine | QABP-90L2, 2.2 kW three-phase induction motor |
| Pole pairs | 2 |
| Single-side air gap | 0.5 mm |
| Bearings | SKF 6205-2Z-C3 (drive and non-drive end) |
| Bearing geometry | 9 balls, 7.94 mm ball, 39.04 mm pitch |
| Bearing orders | BPFO 3.585, BPFI 5.415, BSF 2.357 |
| Drivetrain | torque sensor, two-stage parallel gearbox, magnetic powder brake |
| Vibration | triaxial accelerometer, 100 mV/g |
| Current | three current clamps, 100 mV/A |
| Torque | S2001 sensor, 100 mV/Nm, $\pm$0.5% F.S. |
| Key-phase | one pulse per revolution |
| Sampling | 12.8 kHz, 8 synchronous channels, 90 s per run |
| Diagnostic states | 24 (healthy, single, and compound) |
| Operating conditions | 12 (2 profiles $\times$ {20, 40} Nm $\times$ {1000, 2000, 3000} rpm) |
| Excitation profiles | speed circulation (constant load), torque circulation (constant speed) |
| Runs in the release | 288 (one per state and condition) |
| Compound runs | 108 over 9 compositions |

### *B. Protocol Difficulty Under Operating-Condition Shift*

Windows of 0.64 s (8192 samples, non-overlapping) were described by 108 time- and frequency-domain features (eight temporal statistics and ten spectral descriptors per channel over the three vibration and three current channels) and classified into the 24 diagnostic states by a 300-tree random forest, with three seeds per configuration. Only the split changes across the

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TABLE II
REFERENCE PROTOCOLS, 24-CLASS ACCURACY (MEAN $\pm$ STANDARD DEVIATION OVER FOLDS AND THREE SEEDS). FEATURES AND CLASSIFIER ARE IDENTICAL THROUGHOUT; ONLY THE SPLIT CHANGES.

| Protocol | Folds | Accuracy |
|---|---|---|
| Random windows (leakage-prone reference) | 1 | $0.975 \pm 0.000$ |
| In-condition (guard-gap temporal split) | 1 | $0.965 \pm 0.000$ |
| Unknown condition (leave-one-out) | 12 | $0.939 \pm 0.024$ |
| Cross-profile | 2 | $0.839 \pm 0.115$ |
| Steady $\rightarrow$ transitional segments | 1 | $0.766 \pm 0.001$ |
| Single source (train on one condition) | 12 | $0.333 \pm 0.104$ |

six protocols compared in Table II: *random windows* splits windows irrespective of recording (a deliberately leakage-prone reference); *in-condition* trains on the first 60% of every recording and tests on the final 30% with a 10% guard gap; *unknown condition* holds out each of the twelve operating conditions in turn; *cross-profile* trains on one excitation profile and tests on the other; *steady $\rightarrow$ transitional* trains on quasi-stationary segments and tests on speed and load ramps, with stationarity labeled from the key-phase-derived speed and the torque channel (used only for constructing this protocol, never as a classifier input); and *single source* inverts the unknown-condition protocol, training on one condition and testing on the remaining eleven.

The same features and classifier span 0.975 to 0.333 across protocols. Holding one condition out of twelve costs only $\sim$2.6 accuracy points relative to the in-condition reference, because the eleven retained conditions bracket the held-out one in speed and load and the model interpolates. Training on a single condition and extrapolating to the other eleven costs over 60 points; paired over the twelve condition folds, the difference between the two directions is 0.606 (paired $t$-test $p = 7.7 \times 10^{-10}$; Wilcoxon $p = 4.9 \times 10^{-4}$). Transfer between excitation profiles is likewise asymmetric *at equal training-set size* (144 runs each): training under variable speed generalizes to variable load (accuracy 0.944), whereas the reverse loses $\sim$21 points (0.734; paired over seeds, $p = 3.3 \times 10^{-5}$). Because this contrast is size matched, it isolates excitation diversity rather than data volume, unlike the single-source contrast, which reduces both. Finally, appending 36 speed-invariant envelope-order descriptors leaves the interpolating protocol unchanged (+0.001 on unknown condition, paired $p = 0.84$) but improves the extrapolating one (+0.047 on single source, paired $p = 8.4 \times 10^{-4}$), consistent with their construction.

Two implications carry into the main study. Methodologically, robustness to operating conditions should be claimed in the extrapolating direction, which is why a compositional evaluation should vary composition and compound-context regimes jointly with unseen operating conditions rather than relying on leave-one-condition-out alone. Practically, excitation diversity during commissioning matters at least as much as data volume.

### C. Why Constituent Detectors Use Mechanism-Specific Features

A constituent-level detector can either see the full multimodal feature bank or be restricted to a curated, mechanism-appropriate family. The following experiment measures what sharing one feature space does in this dataset. Multi-label one-vs-rest random-forest detectors were trained on single-fault windows and applied to all compound windows under three feature views: vibration only, current only, and the concatenation of both.

The modality dissociation was complete: vibration features never detected an electrically expressed constituent (detection rate 0.000 across all eight cross-modal compounds) and current features never detected a mechanical one. Critically, concatenation did not combine the two abilities but destroyed one of them. The winding constituent was detected on 39.3% of compound windows from current features alone and on 0.2% once vibration features were concatenated—a roughly 200-fold collapse—while the mechanical constituent retained 0.072 detection in the shared view. The electrical evidence is therefore present and learnable in its own modality, and is lost to the dominant mechanical signature when both families compete inside one representation. Restricting each detector to physically appropriate descriptors removes that competition by construction.

### D. Empirical Checks on Two Feature-Construction Choices

Two parameters of the order-domain descriptors deserve explicit empirical support rather than appeal to convention.

*Integration bandwidth.* Order-domain descriptors integrate spectral energy within $\pm 0.12$ order of a target order. A fixed absolute window is a constant *relative* window only at the fundamental: at the second harmonic it spans half the relative width, while the physical deviation of a bearing order—rolling-element slip makes the true defect order differ from its kinematic value—scales with the harmonic. To test whether this matters in practice, the normalized band ratio was recomputed at half-widths of 0.06, 0.12, 0.25, and 0.50 order and scored by its ability to separate the corresponding single-fault runs from healthy runs (run-level AUROC, median over windows, 84 runs, order resolution 0.017).

The measured peak offsets confirm the concern and bound it. The local maximum of the envelope-order spectrum lies a median 0.024 order from the kinematic BPFO and 0.036 order from BPFI at the fundamental, but 0.049 and 0.080 order at the second harmonic—the offset roughly doubles with harmonic number, as slip-induced deviation should. Consequently the $\pm 0.12$ window contains the observed peak in 100% and 96% of runs for BPFO (first and second harmonic) but in only 96% and 74% of runs for BPFI. The second-harmonic BPFI descriptor is therefore the least reliable member of the bearing family.

Widening the window, however, does not repair this and measurably harms discrimination. At the fundamental, BPFO separates faulty from healthy runs perfectly (AUROC 1.000) at every half-width, so the choice is immaterial there. At the second harmonic, AUROC for BPFO is 0.877 at $\pm 0.12$ but falls to 0.573 at $\pm 0.25$ and to chance (0.497) at $\pm 0.50$. The reason is structural: $2\times$BPFO $= 7.17$ and $2\times$BPFI $= 10.83$ sit close to integer shaft orders, which carry rotational and imbalance energy in healthy and faulty machines alike, so a wider window admits common-mode energy faster than it recovers the displaced fault peak. The $\pm 0.12$ choice is thus close to the best available compromise for this rig rather than an arbitrary one, with the caveat that second-harmonic descriptors—BPFI's in particular—carry less reliable evidence than their fundamentals. A harmonic-scaled window would be the principled alternative, and is left to future work because widening at fixed center is demonstrably not the fix.

*Broken-bar sideband placement.* The current-domain descriptors are evaluated at $o_e \pm 1$ and $o_e \pm 2$, that is, at offsets of one and two *shaft* orders from the electrical carrier. The classical broken-bar signature instead lies at $(1 \pm 2s)f_e$, an offset of $2s\, o_e$ shaft orders, where the slip $s$ follows from the measured carrier and the key-phase speed as $s = 1 - p/o_e$.

The pole count was determined from the data rather than assumed: the detected carrier sits at $50.06$ Hz against a $49.69$ Hz shaft at the 3000 rpm setting (and at $16.63$ Hz against $16.32$ Hz at 1000 rpm), so the machine is two-pole, $p = 1$, and the recovered slip is 3.0%, 1.5%, and 1.0% at the 1000, 2000, and 3000 rpm settings—a near-constant slip *speed* of about $0.3$ Hz, as a roughly constant load torque implies.

The consequence is quantitative. The broken-bar sidebands lie $2s\, o_e = 0.062$, $0.031$, and $0.020$ order from the carrier at the three speed settings (median $0.024$ over 245 analyzed windows), which is only about two spectral bins at the 2-s window length used here, and falls *inside* the $\pm 0.12$ carrier band in 92.7% of windows. The classical broken-bar signature is therefore absorbed into the carrier descriptor itself rather than resolved as a separate feature. The $o_e \pm 1$ and $o_e \pm 2$ descriptors, by contrast, sit one and two shaft orders from the carrier, which is 16 to 50 times further out than the true sideband offset and coincides instead with the $f_e \pm f_r$ spacing associated with eccentricity. These descriptors should accordingly be read as carrier-relative modulation proxies and not as slip-parameterized broken-bar measurements. Resolving genuine broken-bar sidebands on this rig would require either substantially longer analysis windows or explicit slip-tracked demodulation, and remains open.

## References

[1] S. Chen, Z. Liu, C. Li, D. Zou, X. He, and D. Zhou, "Multi-mode fault diagnosis datasets of three-phase asynchronous motor under variable working conditions," *Data in Brief*, vol. 65, p. 112583, 2026.